

# HEALTHCARE DATA CLIENT

## Data Model & DAX Measures

Power BI Data Quality and Exception Reporting

| Document Control    | Detail                                                                          |
|---------------------|---------------------------------------------------------------------------------|
| Engagement          | Jul-Oct 2026                                                                    |
| Role                | Data Analyst                                                                    |
| Primary Stakeholder | Healthcare Analyst                                                              |
| Reporting Tool      | Power BI                                                                        |
| Batch Scale         | Approximately 250,000 records per batch                                         |
| Confirmed Results   | 2% exception rate; review turnaround reduced from 6 hours to 2.5 hours per file |

**Model status:** The reporting concepts, KPI logic and record-versus-failure grain are evidence-based. Exact historical table names, relationships, column names and original DAX expressions are not retained. The model and DAX below are a portfolio reconstruction specification, not a claim of verbatim production code.

# 1. Modeling Objective

Design a Power BI model that supports record-level data-quality KPIs and rule-level exception analysis without double-counting healthcare records that fail more than one validation rule.

## 2. Recommended Logical Model

| Logical Table          | Grain                                                              | Purpose                                                                      | Evidence Status                                                 |
|------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Processed Records      | One row per governed processed record                              | Supports Records Processed, record-level exception status and batch context. | Portfolio reconstruction based on confirmed KPI grain           |
| Validation Failures    | One row per processed record per failed validation rule            | Supports reason-code and multi-failure analysis.                             | Portfolio reconstruction based on confirmed multi-failure logic |
| Validation Rules       | One row per governed validation rule                               | Stores rule category and description for reporting.                          | Portfolio reconstruction based on confirmed rule catalogue      |
| Batch / Source Context | One row per governed processing context where available            | Supports batch reconciliation and filtering.                                 | Portfolio reconstruction                                        |
| Healthcare Domain      | One row per source-supported domain classification where available | Supports domain filtering only when source data contains the classification. | Optional portfolio dimension                                    |

**Key design control:** Processed Records and Validation Failures must remain at different grains. Record-level KPIs should come from the Processed Records table, while rule-level analysis should come from Validation Failures.

## 3. Relationship Design

| From                   | To                  | Cardinality                  | Purpose                                                                |
|------------------------|---------------------|------------------------------|------------------------------------------------------------------------|
| Processed Records      | Validation Failures | One to many                  | Allow one processed record to have zero, one or multiple failed rules. |
| Validation Rules       | Validation Failures | One to many                  | Group failures by rule or reason category.                             |
| Batch / Source Context | Processed Records   | One to many where applicable | Filter processed populations by governed batch context.                |
| Healthcare Domain      | Processed Records   | One to many where supported  | Filter by confirmed domain without inventing domain attribution.       |

**Historical boundary:** The exact physical schema, relationship directions and source keys used in the client implementation are not retained. These relationships define the safest portfolio model.

## 4. Grain Rules

| Measure Type         | Required Grain         | Control                                                                     |
|----------------------|------------------------|-----------------------------------------------------------------------------|
| Records Processed    | Processed record grain | Count the governed processed population once per record.                    |
| Exception Records    | Processed record grain | Count distinct records with one or more failures.                           |
| Exception Rate       | Processed record grain | Use Exception Records divided by Records Processed.                         |
| Validation Failures  | Rule-level grain       | Count individual rule failures.                                             |
| Exceptions by Reason | Rule-level grain       | Use governed reason or rule detail without replacing the record-level rate. |

## 5. Core DAX Measure Specifications

**DAX status:** The expressions below are portfolio implementation examples based on the governed KPI definitions. Because the original table and column names are not retained, they are not represented as historical production DAX.

### Measure 1: Records Processed

Records Processed =  
COUNTROWS ( 'Processed Records' )

### Measure 2: Exception Records

Exception Records =  
CALCULATE (   
DISTINCTCOUNT ( 'Processed Records'[Record Reference] ),  
'Processed Records'[Exception Flag] = TRUE ()  
)

### Measure 3: Exception Rate

Exception Rate =  
DIVIDE ( [Exception Records], [Records Processed] )

### Measure 4: Validation Failures

Validation Failures =  
COUNTROWS ( 'Validation Failures' )

### Measure 5: Non-Exception Records

Non-Exception Records =  
[Records Processed] - [Exception Records]

**Observed result:** The engagement surfaced a 2% Exception Rate. The DAX measure calculates the rate from the governed populations; it must not hard-code 2%.

## 6. Turnaround Measures

### Measure 6: Review Time Reduction

Review Time Reduction Hours =  
[Baseline Review Hours] - [Post Solution Review Hours]

### Measure 7: Review Turnaround Reduction %

Review Turnaround Reduction % =  
DIVIDE ( [Review Time Reduction Hours], [Baseline Review Hours] )

| Measure                       | Confirmed Value / Logic      |
|-------------------------------|------------------------------|
| Baseline Review Hours         | 6 hours per file             |
| Post Solution Review Hours    | 2.5 hours per file           |
| Review Time Reduction         | 3.5 hours per file           |
| Review Turnaround Reduction % | Approximately 58.3%, derived |

**Outcome boundary:** The 58.3% figure is a turnaround reduction only. It is not a productivity, labor-cost, clinical-quality or compliance metric.

## 7. Exception Analysis Measures

### Measure 8: Distinct Exception Records by Current Filter Context

```
Exception Records by Context =  
CALCULATE (  
    DISTINCTCOUNT ( 'Validation Failures'[Record Reference] )  
)
```

### Measure 9: Failure Share

```
Failure Share =  
DIVIDE (  
    [Validation Failures],  
    CALCULATE ( [Validation Failures], ALL ( 'Validation Rules' ) )  
)
```

**Usage boundary:** Failure Share can support a portfolio dashboard when rule-level data exists, but no historical per-rule percentages are retained. Do not populate the visual with invented values.

## 8. Model Validation Controls

| Control              | Expected Result                                                          |
|----------------------|--------------------------------------------------------------------------|
| Processed population | Records Processed reconciles to the governed batch population.           |
| Exception population | Exception Records is a subset of Records Processed.                      |
| Multi-failure record | A record with multiple rule failures counts once in Exception Records.   |
| Failure count        | Validation Failures can exceed Exception Records.                        |
| Exception Rate       | Recalculates as Exception Records / Records Processed.                   |
| Filter context       | Record-level measures remain valid under supported filters.              |
| Rule analysis        | Reason and rule visuals reconcile to Validation Failures.                |
| Turnaround           | 6 hours to 2.5 hours yields 3.5 hours and approximately 58.3% reduction. |

## 9. Reporting Field Usage

| Field Class               | Use in Model                                   |
|---------------------------|------------------------------------------------|
| Record reference          | Distinct record counting and traceability.     |
| Exception flag            | Record-level exception classification.         |
| Validation rule           | Rule-level failure analysis.                   |
| Reason code / explanation | Reviewer-facing exception classification.      |
| Batch reference           | Processing context and reconciliation.         |
| Healthcare domain         | Optional filtering only when source-supported. |

## 10. Design Rules for Public Portfolio Use

| Rule                      | Requirement                                                                       |
|---------------------------|-----------------------------------------------------------------------------------|
| No PHI                    | Do not use real patient, claim, provider or confidential client values.           |
| No invented schema        | Do not present reconstructed table or column names as the original client schema. |
| No invented distributions | Do not create fake exception counts by rule, domain, severity or period.          |
| No hard-coded 2%          | Calculate Exception Rate from the modeled populations.                            |
| No double counting        | Keep record-level measures separate from rule-level failure measures.             |
| Consistent terminology    | Use the same KPI definitions as the BRD, KPI Catalogue and Exception Analysis.    |

## 11. Traceability

| Model/ Measure            | Upstream Artifact                               | Downstream Use                    |
|---------------------------|-------------------------------------------------|-----------------------------------|
| Processed Records grain   | Data Dictionary; KPI Catalogue                  | Power BI Executive Overview       |
| Validation Failures grain | Data Quality Rule Catalogue; Exception Analysis | Reason-code analysis              |
| Exception Records         | KPI Catalogue                                   | Dashboard; Testing                |
| Exception Rate            | KPI Catalogue; BRD                              | Dashboard; Findings               |
| Turnaround measures       | KPI Catalogue                                   | Dashboard; Final Report           |
| DAX examples              | Power BI Requirements                           | Testing; portfolio implementation |